

Vishesh Raju

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Software Development Engineer candidate skilled in distributed systems, cloud-native engineering (AWS), and robotics simulation with hands-on experience in ML-powered perception and scalable backend services.

TECHNICAL SKILLS

- **Languages:** Java, Python, C++, JavaScript, SQL
- **Frameworks & Libraries:** Node.js, Express, React, Flask, PyTorch, TensorFlow, Pandas, NumPy, OpenCV
- **Cloud & Big Data:** AWS Lambda, EC2, S3, Hadoop, Docker, Apache Oozie, MongoDB
- **Web Technologies:** REST API, Microservices Architecture, Distributed Systems, HTML5, CSS3
- **Developer Tools:** Git, GitHub Actions (CI/CD), Jupyter

EDUCATION

New Jersey Institute of Technology (NJIT)

Newark, NJ

Master of Science in Computer Science

Aug 2024 – May 2026

- Current GPA: **3.9/4.0**
- Relevant Coursework: Data Structures and Algorithms, Machine Learning, Big Data, Database System Design, Deep Learning, Operating Systems, Web Development

Expected Graduation: May 2026

R N Shetty Institute of Technology

Bengaluru, India

Bachelor of Engineering in Artificial Intelligence and Machine Learning

Dec 2020 – May 2024

- Cumulative GPA: **7.44/10.0**

EXPERIENCE

Research Assistant

May 2025 – Present

New Jersey Institute of Technology - CIBM3

Newark, NJ

- Engineered scalable Python-based services for automated 3D microscopy data processing, reducing manual effort by **80%** through robust pipeline automation.
- Re-architected the **VAMPIRE microglial analysis pipeline** into modular, production-ready components, improving system reliability and classification performance by **15%**.
- Designed and optimized distributed batch-processing workflows for datasets **200+ GB**, achieving a **2.3x** improvement in throughput using parallelization and efficient memory management.
- Implemented CI/CD-friendly code structure, version control best practices, and reproducible environments to support collaborative development and scalable deployment.

PROJECTS

Autonomous Fleet Simulator

Technologies: Python (FastAPI), React, A Pathfinding, REST APIs*

- Designed an autonomous warehouse fleet simulator with intelligent AMR job routing using Manhattan distance.
- Implemented obstacle-aware A* navigation, smart charging logic, and runtime-editable environments.
- Built a real-time React dashboard with path visualization, live metrics, and job-level analytics.

OS Scheduling and Deadlock Avoidance Simulator

Technologies: Python, React, Flask, REST API, GitHub API

- Developed a full-stack OS simulator with FCFS, SJF, SRTF, Priority, RR scheduling, and Banker's Algorithm.
- Implemented a FastAPI backend with optimized scheduling, resource allocation, validation, and auto API docs.
- Created an interactive React UI with dynamic Gantt charts, editable process tables, and real-time simulations.

Hadoop + Oozie Chicago Crime Analysis

Technologies: Java, Hadoop, Oozie, MapReduce, Python, AWS

- Processed over **8 million records** from the Chicago Crime Dataset using Hadoop and Oozie workflows.
- Developed and optimized **three MapReduce jobs** for crime trend and hotspot analysis.
- Deployed and managed up to **12 AWS virtual machines** for distributed data processing.